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What impact has Concentrating Solar Power (CSP) had on the renewable energy market?

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Abstract

This literature review explores the impact of Concentrating Solar Power (CSP) on the renewable energy market, with a comparative analysis against its direct competitor in the solar energy market: Photovoltaics (PV). The aim is to assess whether CSP's impact is in line with what was predicted a decade ago and investigate the causes of its current status. CSP and PV exhibit significant disparities in adoption and impact due to economic, technical, and socio-political factors.

The review first identifies the Levelized Cost of Electricity (LCOE) as the key economic metric underpinning CSP's challenges. The main findings show how CSP's LCOE declined significantly but has plateaued at approximately \$0.12/kWh since the COVID pandemic. On the other hand, PV's LCOE fell sharply, outpacing CSP and reaching a level of \$0.044/kWh. Furthermore, some technical constraints have been identified as hindrances to CSP's deployment. Site suitability, majorly influenced by Direct Normal Irradiation and slope values, plays a crucial role in the exclusion of lands from potential CSP sites. High water usage and scalability limitations further contribute to promote PV technologies as they are more modular and adaptable to different geographical scenarios. However, thermal energy storage has proven to be a strong advantage of CSP technologies on PV, allowing it to solve the major issue of renewable energy dispatchability, but not enough to close the market gap between the two technologies or gain deployment momentum.

Lastly, socio-political factors strongly influence the deployment of solar plants (both PV and CSP). More specifically, the public's sentiment is majorly driven by familiarity with the technology, the perceived fairness of the project, value and risks, and sustainability matters. While CSP's niche image and large land requirements favour scepticism for its widespread adoption, PV has benefitted from its modularity and adaptability. The political actors have also gradually preferred PV technologies as they are cheaper and more popular, implementing strong funding campaigns and developing policies in PV's favour. This started a self-sustaining loop where PV keeps advancing and CSP stalls.

However, hybrid systems that integrate CSP and PV could allow to reduce costs and enhance performance, suggesting a complementary rather than competitive role for CSP in the future renewable energy landscape.

In conclusion, CSP's minimal market impact is attributed to economic stagnation, technical barriers, and limited social acceptance. Hybridization presents a path forward, positioning CSP as a critical player in enhancing grid stability and meeting long-term energy needs.

Statement of project objectives

The objectives for this literature review are as follows:

- Preliminary knowledge: develop a comprehensive foundation to understand the current state and how it differs from the forecast state of the Concentrating Solar Power technology's impact on the Renewable Energy market.
- Technologies classification: systematically presents the technologies involved (CSP and Photovoltaic), the main systems on the market, and the basic principles and differences between the two. Structure the research to assess the criteria chosen for each of these two technologies.
- Assessment of opportunities: evaluate the causes of the current state of CSP and critically assess the reasons (if any) for the discrepancies with PV. Research and assess future applications and opportunities for CSP, and critically comment on the predictions.
- Referencing convention: this review employs IEEE referencing style adapted to show DOI and ISBN when appropriate (DOIs are shown as hyperlinks).

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Introduction

One decade ago, the International Energy Agency predicted that CSP would produce about 30% as much energy as PV in the following years [1]. However, in 2023 PV dominated the sector with a share of 36.5% of the total global installed renewable power capacity against the 0.2% share of CSP [2]. This indicates that some (or perhaps many) factors were not considered in the initial predictions but turned out to be crucial for implementing and deploying the CSP technologies.

Concentrating Solar Power (CSP) is based on the conversion of solar energy into heat, which is then used to produce steam for electricity generation. The leading country is Spain with 2300 MW of installed capacity, followed by the USA and China. The commonly deployed CSP systems are categorized into two main types: linear concentrators and point concentrators.

Linear Concentrators:

- I. Parabolic Trough Collector (PTC): Parabolic trough collectors use long, curved mirrors to concentrate sunlight onto a receiver pipe containing a heat-transfer fluid, such as oil or molten salt, heated to 350–550°C. The heated fluid transfers energy to a heat exchanger, generating steam to drive a turbine and produce electricity, achieving solar-to-electric efficiency of around 15% [3], [4] and thermal efficiencies of 60-80% [4]. Equipped with thermal energy storage, often with two-tank molten salt systems, PTC systems can continue generating power even during periods without sunlight. They are efficient in land use and are among the most advanced and widely used CSP technologies.
- II. Linear Fresnel: Linear Fresnel systems use parallel flat or slightly curved mirror strips to focus sunlight onto a stationary receiver mounted above the mirrors [4]. Water is heated within the receiver tubes to produce steam for turbine-driven power generation. This configuration reduces costs due to simpler reflectors but achieves lower solar-to-electric efficiencies of 8–10% [3], making it more suitable for smaller-scale applications.

Point Concentrators:

- III. Solar Parabolic Dish (SPD): SPD systems use dish-shaped mirrors to concentrate sunlight onto a receiver at the focal point, where the heat drives a Stirling or Brayton engine for electricity generation [5]. These systems achieve the highest solar-to-electric efficiencies among CSP technologies, at 25–35%, and are highly effective due to precise solar tracking and minimal optical losses. Concentrators are typically made of glass or plastic surfaces coated with reflective materials like silver, achieving high reflectance

values. A single dish can generate up to 0.5 MW making them ideal for off-grid and decentralized energy applications [3].

- IV. Central Receiver (or Solar Power Tower - SPT): Central receiver systems use a field of heliostats (computer-controlled mirrors) to reflect sunlight onto a receiver mounted on a central tower. The receiver heats a working fluid, often molten salt, water/steam, or air, to temperatures exceeding 530°C. Molten salt is a preferred working fluid due to its excellent heat storage capacity, thermal stability, and non-flammable properties, allowing power generation for up to 13 hours after sunset. The heat is used to produce steam, which drives a turbine to generate electricity. These systems also achieve solar-to-electric efficiencies of 20–35% [3] and innovations in tracking systems, mirror cleanliness, and optical performance constantly enhance efficiency. They have high land requirements, but are ideal for large-scale plants, accounting for 75% of the installed plants [6].

So, the solar field in a CSP plant captures and converts sunlight into thermal energy, which is transferred to the power block for steam generation. A critical design parameter is the solar multiple: defined as the ratio of the collector area to the area required to produce the rated power output under standard conditions, indicating how much additional thermal energy can be captured for storage or off-peak operation. Another crucial factor is DNI (Direct Normal Irradiation): the measure of solar radiation received per unit area by a surface perpendicular to the sun's rays, excluding then diffuse and reflected radiation. Additionally, thermal energy storage systems, particularly two-tank molten salt, are fundamental for continuous electricity generation and grid stability. The average lifetime for CSP plants is 20 to 30 years [7], [8].

On the other hand, Photovoltaic (PV) devices convert sunlight directly into electrical energy according to the photovoltaic phenomenon. The smallest unit is called “cell”; it is made of semiconductor materials - such as silicon - and coated by layers of glass and plastic to be resistant. Cells are interconnected to form the “modules” which are then connected to form “arrays” that may be connected directly to the grid or can be combined with other arrays connected to the grid to form the PV systems. Several technologies can be used to produce PV energy. Crystalline silicon (c-Si) and thin-film technologies dominate the market, with c-Si accounting for 95% of the worldwide share of production [9], [10].

The main indicator for the economic analysis is the Levelized Cost of Electricity (LCOE) [11] which is used to compare the costs of different energy technologies. However, research shows that the market impact of CSP and PV is also driven by other factors that have influenced the development and deployment of the two technologies.

In this literature review, the main barriers to CSP adoption as the primary solar source of energy are investigated and the potential opportunities of the technology are assessed in order to understand whether PV is the cause of CSP's downfall or if the limitations are solely dependent on the technology and its implications.

The analysis will first dress the economic situation and issues, then examine the social acceptance and political barriers to the technology. To better adhere to the scope of the literature research, the comparison will treat the purely technological aspects relatively superficially, to provide essential knowledge while keeping the focus on the key factors that differentiate CSP and PV market impact.

1 – The economic situation

The key factor identified in the comparison between CSP and PV technologies is the economic viability of the plants, making it a sensible starting point for analysis. The metric used to compare different Renewable Energy technologies from an economic point of view is the LCOE. In general, the LCOE is the ratio of the Lifecycle Cost to the Lifetime Energy Production [12]. The LCOE can be calculated using several different formulas to

encapsulate different costs and benefits of different technologies. Khan et al. [8] provide a comprehensive review of the most employed approaches in the literature for CSP.

As illustrated in Figure 1, the LCOEs for both CSP (a) and PV (b) plants declined significantly in the past decade. More specifically, LCOE for CSP fell by 70% during the 2010-2023 period, and PV's LCOE decreased by 90% over the same timeframe [13].

1.1 Concentrated Solar Power LCOE:

The most significant parameters encapsulated by the CSP's LCOE are the plant's capacity factor and efficiency, the DNI at the operating site, the operation and maintenance of annual operation (including insurance), the initial investment (site development, components and system, assembly, grid connection, and financing), and the plant's economic lifetime [14], [15]. The construction and maintenance of large-scale CSP plants require substantial initial capital investments, which negatively impact the overall project's economic viability as they represent a major upfront cost – marginally impacted by discount rates – that has to be

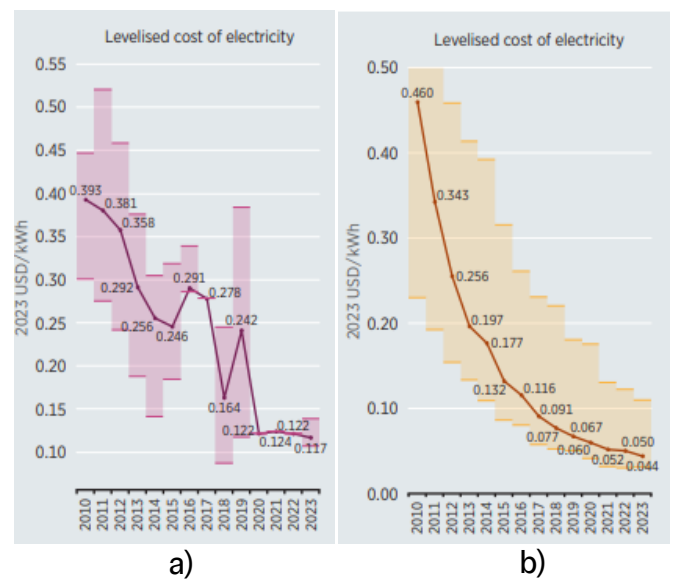
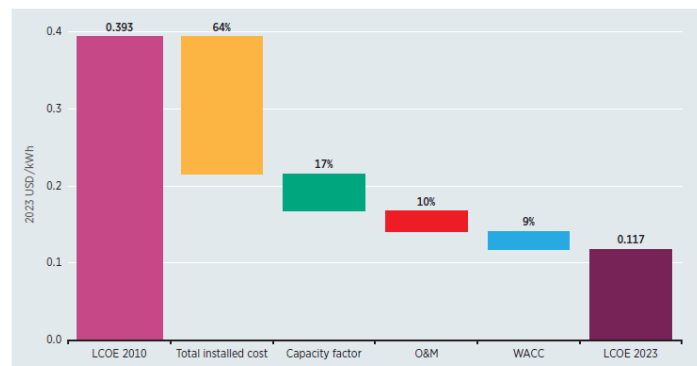


Figure 1: a) Levelised Cost of Electricity for CSP plants from 2010 to 2023. Reprinted from [13] b) Levelised Cost of Electricity for PV plants from 2010 to 2023. Reprinted from [13]

balanced by future earnings – majorly impacted by discount rates. The effect of discount rates can be seen in the economic indicator analysis by Colla, Ioannou [12]. Additionally, the complexity of CSP systems, particularly in the case of PTC and molten salt storage, further contributes to elevated operational and maintenance costs.

Figure 2 presents the breakdown of the LCOE reduction for CSP, highlighting that the primary contributor to this decline is the reduction in “total installed cost”, accounting for 64% of the drop. This contributor includes advancements in technology and cost reductions associated with thermal energy storage improvements [13].

When analysing the LCOE for CSP, it is crucial to account for two factors that strongly influence the data: the LCOE is extremely "sub-technology specific" (i.e., LCOEs for Parabolic Trough Collectors and Solar Power Towers differ) and strongly country-specific. These characteristics make it challenging to generalise. As PTC technology is the most widely deployed in terms of capacity (as shown in



Notes: kWh = kilowatt hour; LCOE = levelised cost of electricity; O&M = operation and maintenance; WACC = weighted average cost of capital.

Figure 2: LCOE reduction for CSP per category in the period 2010 - 2023. Reprinted from [13]

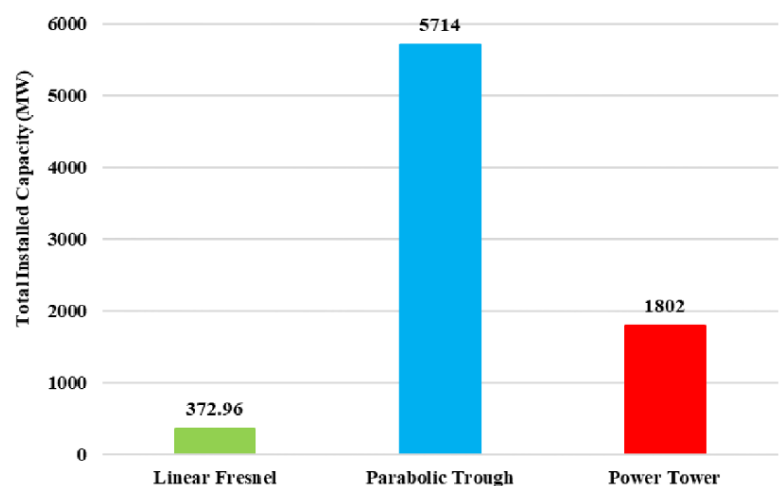


Figure 3: Installed capacity of CSP technologies by technology as of 2023. Reprinted from [6]

Figure 3) followed by Power Tower plants [6], it is reasonable to use a general LCOE – as it is commonly done in the literature – that likely reflects the LCOE of PTC systems. The issue of country specificity is evident if data from the 2010–2012 period – when Spain dominated the CSP market [16] – and from 2020 – when China completed the only two CSP projects of the year – is analysed and compared to the world data, to notice how the country-specificity of deployed plants drives the LCOE [13]. Thus, it is crucial that any conclusions drawn from a specific case study are contextualized, with appropriate considerations made when attempting to generalise the results.

1.2 Photovoltaic LCOE:

Photovoltaic technology has experienced a substantial reduction in costs over the years, largely attributed to economies of scale, technological advancements, and improvements in manufacturing efficiency as will be explored later. Similar to CSP's LCOE, the LCOE for PV incorporates factors such as initial investment, the plant's capacity factor and efficiency, system degradation, annual operation and maintenance costs (including insurance), capital costs, and the system's operational lifetime [15], [17].

The breakdown of the LCOE reduction for PV is presented in Figure 4, demonstrating that the primary contributor to this decline is the reduction in module costs [13]. This is a direct consequence of the extensive investment campaigns aimed at reducing PV LCOE, such as the U.S. Department of Energy's funding program to

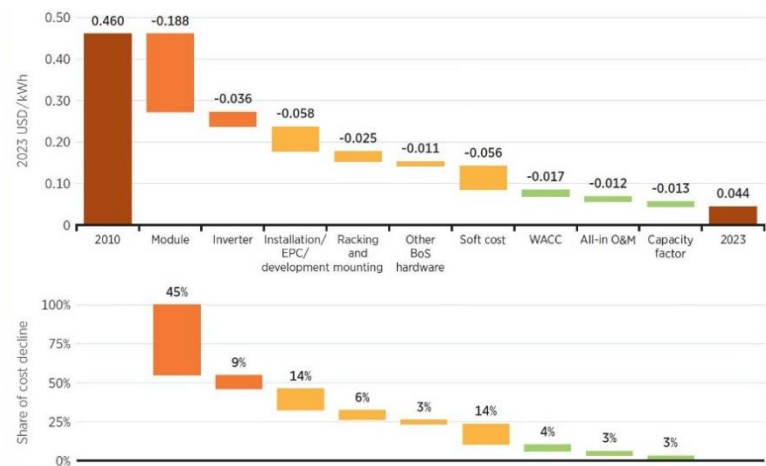


Figure 4: Photovoltaics LCOE reduction with category breakdown in the period 2010-2023. Reprinted from [13]

achieve the SunShot 2030 goals [18]. An important factor is the module's strong dependence on commodities and material costs [19] and since crystalline silicon-based technologies have accounted for over 90% of PV technologies deployed since 2012 [9], [10], it is reasonable to state that the PV costs are heavily influenced by material expenses. Moreover, most of the module production occurs in Asia, particularly in China [20], introducing a critical dimension to the discussion: the influence on PV module costs (and on the LCOE) of the export costs, tariffs, and routes for worldwide technology employment.

1.3 Economic comparison:

The economic competition between CSP and PV is largely defined by their respective LCOE trajectories and the market dynamics these have fostered. While both technologies have seen significant cost reductions over the past decade – CSP's LCOE declining by 70% and PV's by 90% [13] – the recent economic history highlights a critical divergence. Over the past three years, CSP's LCOE has plateaued at approximately 0.12 USD/kWh, while PV's LCOE has dropped by over 25%, now standing at less than half of CSP's LCOE, highlighting PV's ability to rapidly progress and its growing dominance in the renewable energy sector.

PV's superior economic momentum is a direct consequence of its self-reinforcing cycle: being a more convenient and broadly applicable technology, it has garnered greater deployment (63% added PV capacity in 2021-2023, only 8% for CSP [21]), which in turn has attracted more investment and funding for R&D, leading to further cost reductions. This loop has enabled PV's LCOE to drop from being 117% of CSP's in 2010 to less than 38% by 2023 [13]. In contrast, CSP's progress has been hampered by its higher initial costs and dependency on site-specific conditions like DNI requirements [22]. This dynamic presented a critical challenge for CSP since it hadn't been able to identify and demonstrate advantages in areas where PV cannot compete – such as its potential for long-duration energy storage and dispatchable solar power [23]. This feature could make CSP appealing to investors by emphasizing a long-term value proposition that justifies its higher costs. Without this, the economic gap between CSP and PV will likely continue to widen, reinforcing PV's dominance in the solar (and renewable) energy market (Figure 5 shows the predicted 2030 energy outlook according to [24]).

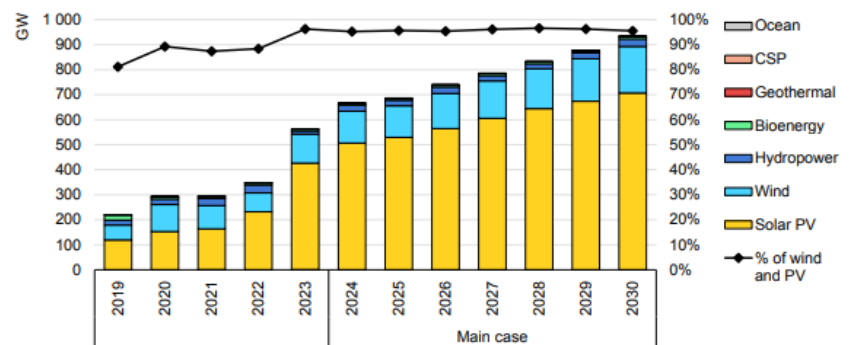


Figure 5: Projections of renewable energy capacity until 2030 by technology. Reprinted from [24].

2 – The technical differences

Previously, the LCOE analysis suggested that the cost's behaviour strictly follows technological development. As such, it can be valuable to investigate the literature that explores the technical and technological discrepancies in the two technologies as these contribute to the economic discrepancy, which is in turn one of the causes of the stumbling of CSP compared to PV.

2.1 Land suitability & site selection:

One of the possible causes of the CSP's low deployment capacity is the strict site requirements. Understanding the reasons for the differences in location suitability of the two technologies gives insights into the first discriminatory point between PV and CSP. From the literature analysis, it emerges that most of the models to assess site suitability are based on Geographic Information System (GIS) and Multicriteria Decision Making (MCDM) and are supported by an Analytic Hierarchy Process (AHP). This method is sufficiently accurate and it is applied to case studies of countries in every part of the world: Aly, Jensen [25] proposed this model to determine the most suitable areas in Tanzania for PV and CSP plants; Giamalaki

and Tsoutsos [26] used the model with techno-economic criteria and with socio-environmental criteria, assessing the suitability areas in Crete according to three different scenarios (assigning different weightings to criteria); and Alqaderi, Emar [27] analysed the UAE land suitability for CSP. The analysis method starts by assessing criteria such as protected areas, land cover, topography, water bodies, solar radiation, and distance from cities, roads, and power lines. Several of the potential regions for CSP are effectively discarded due to some of these factors, that makes them significant for the comparison: water availability, topography, and solar radiation. Alqaderi, Emar [27] indicates a consensual maximum of 2.1% of slope and a minimum DNI of 1800 kWh/m² for CSP plants, while for PV there is no consensual maximum slope with ranges from 3% (as used by Uyan [28]) to 11% (as used by Noorollahi, Fadai [29]), implying that a more flexible approach is available for PV plants based on topography. The minimum DNI of 1700 kWh/m² for PV [25] allows several sites to be suitable for this technology but not for CSP: using ESMAP, SOLARGIS [30] data, this criterion can be investigated to see how it strongly discriminates suitability in some areas. Using the south of Italy as example we can see how in the southern regions – more favourable as DNI is higher – the DNI's values are between 1700 and 1800 kWh/m² (Figure 6).

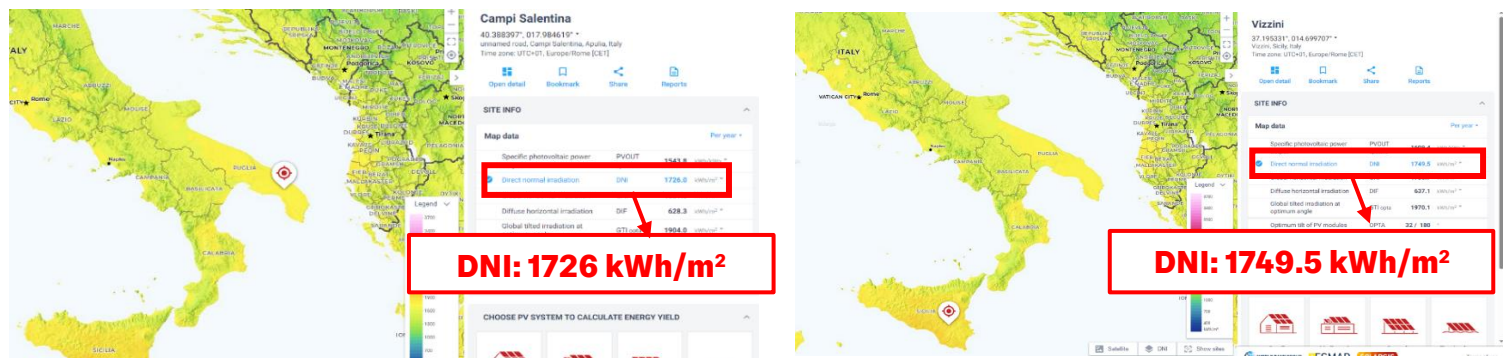


Figure 6: DNI values for two locations in the south of Italy. Screenshots from [30] and details added to enhance readability of the values.

The results of the comparative studies of CSP and PV show how, in general, PV has a less strict land requirement, allowing its deployment in areas where the DNI is not sufficient for CSP to work efficiently but is sufficient for PV. For instance, according to Aly, Jensen [25] a total of roughly 45,000 km² were identified as suitable sites for CSP plants in Tanzania, while 168,000 km² (3 times as much) were indicated as suitable for PV plants [25]. As can be seen in Figure 7, the situation is significantly more flexible for PV, increasing the probability that suitability is met after considering all the other criteria (like slope and GHI). In most of the countries assessed using this model, DNI is one of the strongest criteria for site suitability, limiting significantly the available areas for CSP. Therefore, the land suitability issue plays a key role in the increase of the gap between the PV and CSP deployment.

2.2 Scalability, land use:

Concentrated Solar Power plants require substantial land due to the devices used for the concentration and focus of sunlight – including but not limited to heliostats, tracking systems, balancing systems, cooling and cleaning systems.

Moreover, CSP plants are typically deployed in large-scale installations since cost for the technology is very sensitive to scale. Typically, large-scale power plants (≥ 50 MW) are more cost-effective, minimizing energy production costs which require relatively large capital investments and financial risks – in part because of the great technical complexity [8]. This type and size of plants requires a significant amount of land primarily used for the installation of heliostats or mirrors – as introduced above – which need to be arranged in a way that maximizes solar concentration, and this doesn't always result in adjacent mirrors but may require clearance between components. Compared to PV, CSP on average has always had a larger land usage as shown by Ong, Campbell. [31] over ten years ago. For this reason, CSP plants thrive in regions with ample open land and high solar irradiation but are never the best choice for smaller projects, since PV can be applied to various scenarios thanks to its modularity and scalability without major changes in LCOE (but with a reasonably wide range of capital cost varying with the dimension) [32]. The major issues with CSP land usage are the habitat and soil destruction, and the severe effects on the surrounding environment [33]. However, a strength of the technology – as described by Marzouk [34] – is that CSP can produce more electricity than PV in the same area.

The photovoltaic systems' advantage is that they offer exceptional scalability, and are capable of being deployed in various sizes, ranging from small rooftop installations to large-scale solar farms. Therefore, PV installations are flexible in terms of land use, as they can be integrated into several available surfaces and the size of a PV plant depends on factors such as available space, solar resource potential, electricity demand, and project goals. The literature – for example Abuelrub, Saadeh [35] and Abo-Khalil, Sayed [36] – presents various sizing methods. This versatility makes PV systems suitable for both centralized and distributed power generation.

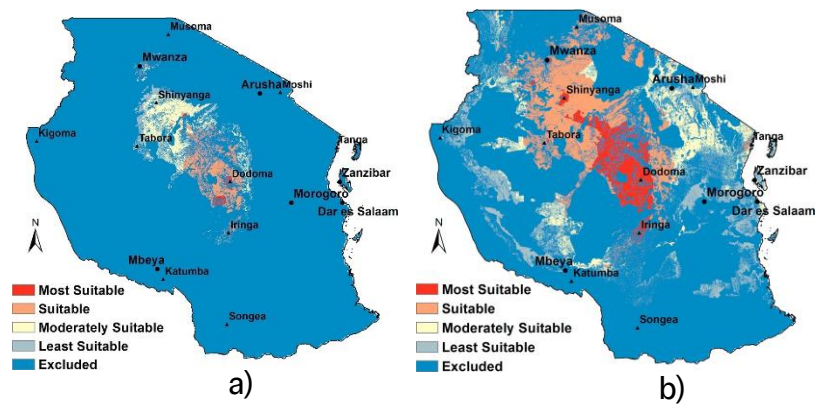


Figure 7: a) Site suitability gradient for CSP in Tanzania. Reprinted from [25]. b) Site suitability gradient for PV in Tanzania. Reprinted from [25].

2.3 Water use:

A pressing issue in renewable energy power plants is water consumption, becoming a strong discriminating criterion for selecting the location (see above), a potential factor of increased costs, and a negative impact on the sustainability and image of the plants which, as seen below, plays a role in the social acceptance.

One of the main drawbacks of the CSP technologies through the development stages in arid regions during the 2010s was the high water consumption. According to the EASAC [37], in 2011 the water consumption for a wet-cooled plant was of 3.6 l/kWh, 90% of which was used for wet cooling and 10% for cleaning. Building on this, Qoaider and Liqreina [38] in 2015 investigated how transitioning to dry cooling can lead to major benefits as the water consumption can reduce significantly. The results show how on a 50 MW plant in Jordan the annual water usage goes from 735,943 m³/a to 62,152 m³/a (a 92% reduction) [38]. Several applications influence the total water usage of a CSP plant: cooling system, steam cycle, mirror cleaning, filter backwash, bundle cleaning, and auxiliary machine cooling for example [39] and the main source of water is usually ground or surface water from natural reservoirs. Therefore, the quality of the water depends directly on the local conditions and, in most cases, needs treatments to achieve a level of “purity” suitable for the different consumers in the plant. CSP plants also produce wastewater in different ways: effluent, concentrate, backwash water, and discharge are continually produced from treatment processes, cleaning, or other processes [39]. According to Mapes and Larsen [40], water consumption in CSP technologies in Spain as a sample country is set to be the largest share of the total water consumption in every possible projected scenario for 2050, highlighting also that CSP has one of the most important water consumptions among all the renewables.

2.4 Energy storage systems:

The most common energy storage system for power grids is pumped hydropower. However, the preferred Energy Storage System (ESS) employed in PV plants are batteries, while for CSP it is Thermal Energy Storage (TES) [41]. This follows logically from the nature of the technologies themselves: PV converts solar energy into electricity thus the batteries are the easiest way to store it, while CSP converts the solar energy into heat that can easily be stored in TES. Chen, Cong [42] presents the fundamentals of the storage technologies used in industry. Batteries are widely used ESS due to their modularity, responsiveness, and adaptability to various applications. There are several types of batteries but the most common are lead-acid, lithium-ion, and flow batteries. Each one is suitable for a specific

application based on the requirements but challenges such as high costs, limited energy density, and environmental concerns related to materials pose major issues for the industry. On the other hand, Thermal Energy Storage is based on storage of heat for later conversion into electricity or for direct application. High-temperature TES – including molten salts or Phase Change Materials – enables integration with high-temperature processes, such as CSP. TES provides cost-effective, long-term storage but lower efficiencies (30–60%) compared to batteries, which are usually more expensive but can reach up to 95% of efficiency. The PV modules produce at peak around midday when the amount of energy demanded is usually exceeded. In off-peak hours, the demand is greater than the generation of the PV plants and could be mitigated by the stored energy from the peak-hour surplus. However, most of the time, the cost of the additional energy storage system is higher than the “loss” due excess electricity and therefore, it isn’t the best solution.

For this reason, ESS is CSP's major advantage over PV. The thermal energy storage systems currently employed are molten salt storage tanks and steam accumulators [6]. Achkari and El Fadar [23] present a list of all the operational and under-development CSP plants in the world up to 2019 from which it's evident that two tank molten salts are the prevalent technology in use.

As opposed to PV, the introduction of TES in systems doesn't necessarily imply that the costs are driven up: according to Khan, Gutiérrez-Alvarez [8] the capital cost decline (shown in Figure 8) continued even if the size of TES increased over time. This presumably allowed to “shift away” from no storage after 2013 – when the market expansion outside of Spain (Figure 9) began [43] – and to build larger plants with TES systems.

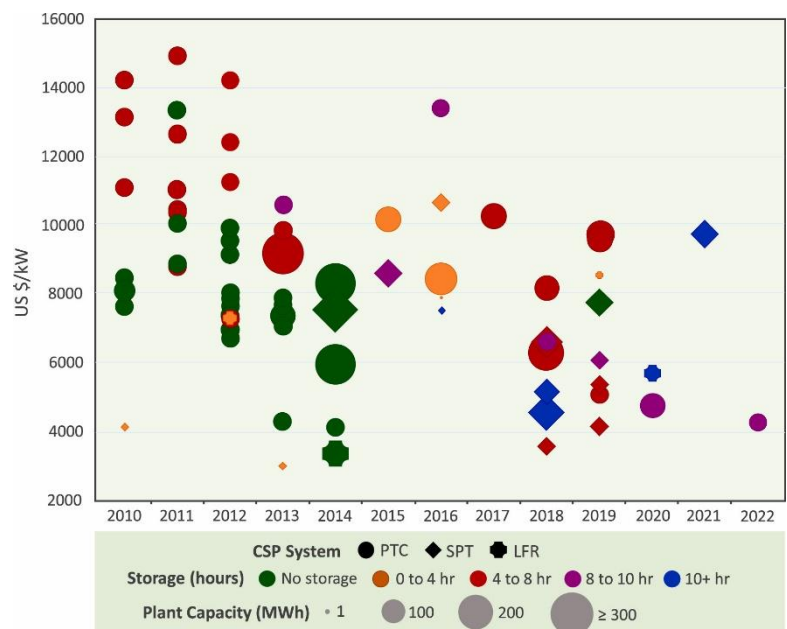


Figure 8: Installed cost of CSP plants by dimension, storage, and system from 2010 to 2022. Reprinted from [8].

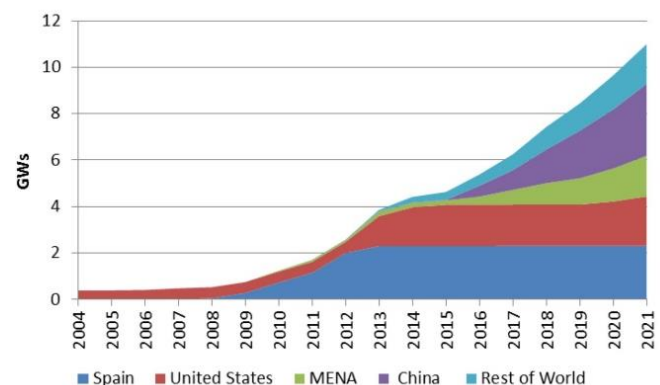


Figure 9: Installed CSP capacity per country or region from 2004 to 2021. Reprinted from [43].

The following question arises naturally then: if CSP gained the great advantage of functional energy storage, which means dispatchability, why is it not competing again with PV? To answer this, it is important to remember that when the CSP market started to implement TES extensively, PV had already created a gap in the LCOE that could hardly be closed by acting on a single factor. Moreover, this technological feature is more of a “grid stabilizer” rather than a standalone technology. Therefore, CSP with TES is not competing against PV, but rather it is competing against the other energy resources with dispatchability such as fossil fuels to be integrated into the grid to enhance dispatchability. This sounds counterintuitive but, up until now (as will be reviewed in section 3), the fossil fuels-based plants were not disincentivised enough or the renewables were still lacking support, and to ensure dispatchability often fossil fuels were paired with renewables. For example, Lingxiang Yao, Zhiwen Guan. [44] studied the feasibility of replacing a grid-connected coal-fired power plant with a CSP plant in China and the possible implications. This explains the apparent miss out of the Energy Storage competition between PV and CSP: the competition is, in reality, between CSP and other technologies to supply energy when PV is not sufficient. [45] highlights this mindset and focuses on a crucial point that emerged from this review: PV and CSP used to compete but now PV clearly prevailed, and the CSP’s competition is shifting from “being the main competitor” to “being the main complement”.

2.5 PV-specific technology improvements:

According to Fischer, Woodhouse. [46], the PV technologies can reduce the module costs by acting on three main aspects:

- Increasing the ratio of module area efficiency improvement to cost increment.
- Piece cost optimisation, including the usage of Si-based material more efficiently.
- Introduce specialised modules for different applications.

The more interesting point from a technological point of view is the cost reduction (past and future) due to the optimization of poly-Si usage, which contributes significantly to the reduction of the PV’s LCOE [13] raising two topics of discussion. Firstly, Hallam, Kim [47] shows the learning curve and the trend of poly-Si used per cell/module and per industry, as well as the requirements to meet the ITPRV targets [46]. It can be seen clearly that the poly-Si consumption decreased substantially in the past years (with increasing cumulative PV installed capacity) tending to the ideal limits. These ideal limits however pose a potential halt signal for the future PV LCOE decline. Secondly, the correlation of the LCOE with the price of Si introduced above is a political, economic, and technical factor that has the potential to

significantly impact the LCOE as demonstrated in the Covid-19 pandemic period. According to the IEA, the price changes of commodities and logistics were the reason of a 30% increase of the LCOE, for the 2020-2022 (Covid) period [48] that had to be attenuated by a 10% reduction in technology improvements as seen in Figure 10.

A further consideration is the modularity of PV power plants: the PV plants are composed of a multitude of arrays, so the economy of scale doesn't fully apply in this situation as the cost of PV modules don't benefit from mass production. This defines a linear relation of variable cost of the plant and number of modules installed.

The US NREL reports that for utility-scale plants this translates into the module cost accounting for 30-40% of the total costs associated with installing photovoltaic (PV) systems for utility-scale ground-mount projects (Figure 11) [49].

3 – The socio-political aspects

3.1 Population acceptance and opinions:

One of the most important factors that determine the success of the implementation of large-scale energy power plant projects is the society's response to the announcement. Society is defined as every actor involved but not directly the commissioner or the deployer. The heterogeneity of actors involved poses a particular issue when investigating this factor: it is clear that the response in the metropolis in California is different from that in rural villages of the Moroccan Shara Desert.

Firstly, Batel, Devine-Wright [50] investigated the literature on this topic to better define what is meant by social "acceptance" and the difference with "support". In this analysis, people's acceptance is defined as a passive response to an external imposition, while support is an active factor in the growth or deployment of the RET plants. To demonstrate it, Batel, Devine-Wright [50] carried out a case study on high-voltage power lines in the UK. Concluding that if there isn't acceptance, there is also no support. However, acceptance

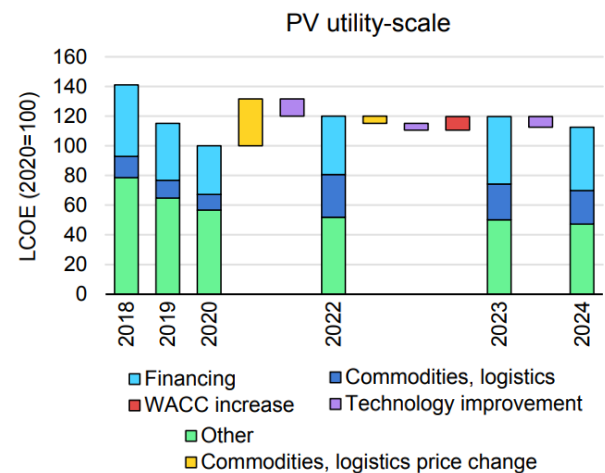


Figure 10: LCOE for PV as a percentage of 2020 LCOE in the period 2018-2024 with category breakdowns. Reprinted and edited from [48].

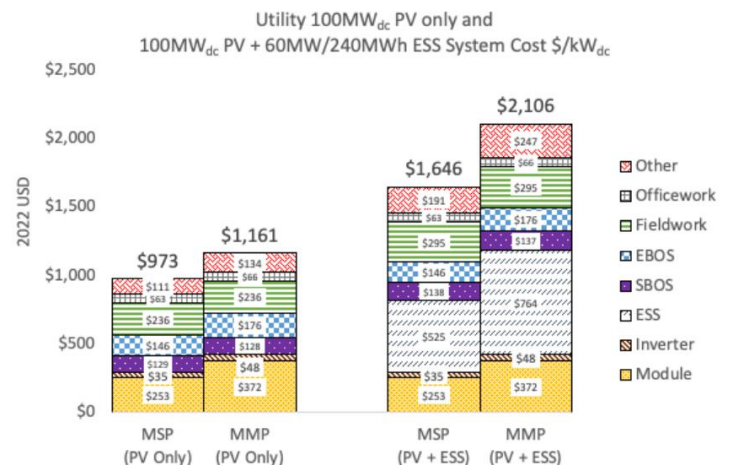


Figure 11: PV costs for 100MW plants with and without ESS divided by category. Reprinted from [49].

doesn't necessarily mean that the project is supported. Although these concepts are defined better, it is quite hard to quantitatively assess social feelings. Therefore, keeping in mind the difference between acceptance and support, the literature research tries to identify the main factors involved and the general acceptance trends (if any) from around the world. In 2018, Huijts, Molin [51] proposed a framework to assess social support and defined three factors that influence it: overall evaluation of costs, risks and benefits, moral evaluations (positive or negative effect on the environment or society for example), and positive or negative feelings related to the technology (like satisfaction, anger, fear, ...).

In addition to these base factors, trust in the actors and fairness of operations are two main causes of positive or negative attitudes towards the technology, specifically when the actors have no experience with the technology or knowledge of how it works [51]. Social acceptance is generally tested using surveys, allowing for a more structured response and comparison with existing studies. To investigate how behaviour changes around the globe, the literature analysed covers very different geographical areas. Yuan, Zuo [52] investigate the social response by applying the method aforementioned to PV and Solar Water Heating – even if it isn't directly CSP, the study remains meaningful to understand the view on the general topic – in urban and rural areas in China. The results of the study show how public awareness is significantly greater in urban areas rather than rural areas, the technologies are better known within educated groups of people (those who completed university) and mainly aged 21-60. However, SWH is more familiar than PV as it is widely deployed in the regions assessed. Regarding PV implementation, 60% of the people interviewed declared themselves to be neutral, and the remaining were similarly split between “agreement” and “disagreement”. The situation shifts significantly towards an agreement for SWH, indicating that the factor of awareness (or “knowledge”) identified by Huijts, Molin [51] does indeed influence the opinion and response of the population. On a similar note, Zoellner, Schweizer-Ries [53] investigated public acceptance of PV, wind, and biomass plants in Germany. For the scope of this research, only the PV results are discussed. The investigation assessed two different areas in Germany, one with an installed PV system and one where the installation of a planned PV system was prevented by hindrances. Some of the factors accounted for by the study are image, agreement (with communities and between stakeholders), communal commitment, location of the plant, financing, and participation of citizens. Not surprisingly, the correlation between perceived fairness and reported acceptance exists, and the participation of the community in the planning process is crucial to gain support. Equally important is the perceived justice: the transparency of implementation processes is relevant

to avoid marginalization and opposition to the project. Segreto, Principe [54] review the European area's response to the topic, where renewable energy projects tend to be easily accepted as a concept, but when it comes to local acceptance – communities directly involved – the support rate falls. The results confirm that the drivers are trust, procedural justice, financial benefits, and socio-demographic factors. However, in the EU multi-country scenario other factors such as perceived rivalries and benefits arise. The paper from Burghard, Dütschke [55] investigates how CSP cooperation mechanisms are perceived in Europe. In this case, a project with Spain as the host country and Germany and the Netherlands as off-taker countries is investigated and a positive response is recorded. However, the complexity of implementation and high costs are noted as significant downsides for cross-border projects in the EU. While most stakeholders in Spain support CSP projects due to local economic benefits and technological leadership, environmental impacts and noted as barriers. CSP is seen as a cost-efficient way to meet renewable energy targets, especially given space constraints in the Netherlands, but national interests and lack of political reliability are major obstacles, so the preference for national strategies over cross-border cooperation that emerges from the study is strong [55]. Hanger, Komendantova [56] studied the social acceptance of the Noor I solar project in Morocco [57] surveying both urban and rural areas. The findings are in line with the models: the most important drivers for local communities are the awareness (“level of information”) and the environmentally friendly image of the project. It is valuable to notice that the surveyed population in this case is not rightfully representative of the actual population as women were underrepresented and misrepresented (83 females and 149 males, women under 35 did not participate). The average level of education and the employment categories differ slightly from the 2004 census representation of the society. The general feedback in the region, however, was positive and driven by a decent level of information about the topic. Regarding the United States, Susskind, Chun [58] analysed the sources of opposition to 53 unsuccessful RE projects in the US and categorized them. The findings show that environmental impacts, land value, public participation, financial feasibility, health and safety, and intergovernmental disputes are the main causes of 4600 MW loss of potential installed capacity in the period from 2008 to 2021. The results from all these studies validate the social acceptance model as a framework for a variety of projects involving large plants associated with renewable energy and requiring great funding and innovative technologies. Therefore, the general idea about solar plants is positive but the consideration decreases with the scale of the project and the vicinity of the plant. Acceptance of CSP or PV isn't

quantified numerically but the more familiar they are, the more they will be accepted. Thus, PV's popularity around the world and its image of scalable and clean energy is opposed to CSP's uncertain appearance as a fairly unknown landscape-damaging clean technology requiring a lot of land.

3.2 Policies and political aspects:

Investigating social attitudes to solar technologies, the policies and the political choices surrounding the topic can give a clear indication of what the governments or the agencies involved want to pursue. For instance, it is interesting to analyse the politics surrounding the Ouarzazate Solar Project in Morocco [57] introduced above. According to Rignall [59], to achieve the goal of reducing dependency on fossil fuels and develop 2000MW of solar energy capacity, the Moroccan government implemented a series of policies that caused issues in the region: the government's acting points resulted in land dispossession, marginalization (excluding locals from the decision-making process), economic inequality, and environmental impact. The three main policies adopted by the government were [59]:

1. Solar Plan: The national initiative to develop solar energy capacity and reduce fossil fuel dependency.
2. Land Acquisition: Legal procedures based on colonial frameworks – colonial laws placed collectively owned lands under state's control – to acquire land for the project, excluding the communities involved from any consultation. Communities made broader political claims about resource access, economic justice, and their right to benefit from national development projects. The problem extends beyond this, and the policy has been perceived as a territorialization (integrating peripheral regions into the national economy by framing them as "marginal" or "underutilized" to justify their exploitation).
3. Technocratic approach and Development Interventions: The government framed popular resistance as a set of technocratic problems to be solved through development interventions to depoliticize the opposition by addressing it through bureaucratic means. At the same time, in response to local resistance, the state implemented community development projects funded by the proceeds from the land sale to placate local demands but often failed to address the root causes of resistance.

As predicted by Huijts, Molin [51], this caused a strong opposition to the project specifically from the citizens that have been dispossessed of land, and those who realized that the project was oriented to favour European countries and Moroccan elite. Frisari and Stadelmann [60] focuses once again on the same project, but on the financial structures, risk

mitigation, and the related policy effectiveness instead of the socio-political impacts and community resistance related policies. More specifically, the policies discussed by Frisari and Stadelmann [60] include specific financial mechanisms like competitive bidding, PPAs, and viability gap funding. The Moroccan government in fact acted to ensure reliability, long term revenue, and mitigate risks. The move included a two-stage competitive bidding – where bidders first qualify based on technical and financial criteria and then compete on project costs –, a long-term power purchase agreement (PPA) and guaranteed offtake – consisting of the MASEN (Moroccan Agency for Solar Energy) entering into a 25-year PPA at the auction price, guaranteeing the purchase of solar power –, and viability gap funding guarantees – the Moroccan government guarantees MASEN's financial viability, supported by international financial institutions (IFIs) with concessional loans and grants. The same paper focuses also on the policies to financially promote and de-risk the National Solar Mission (NSM) and its framework in India. Main policies include reverse auction/competitive bidding to procure solar power cost-effectively having developers to bid a discount to a reference tariff (with the lowest bids winning), and a 25-year PPA with a guaranteed offtake similarly to Morocco [60].

3.3 Brief environmental outlook:

Comparing CSP to PV, it can generally be said that the two technologies are limited by similar environmental setbacks and GHG emissions. For PV, the manufacturing processes of panels are relatively energy-intensive [61] but have become more and more sustainable thanks to the developments in the use of clean energy sources in the production phase and of the advancements in recycling. On the other hand, CSP systems – particularly when the ESS is molten salts – present environmental challenges mainly due to hazardous materials and land disruptions [6]. Specifically, the large land requirements for CSP plants – particularly the power tower configurations – have a strong impact on local ecosystems endangering complete areas and the respective wildlife [33]. The impacts can be mitigated only through a detailed and proper site selection, great water management, and constant R&D advancements targeted specifically at sustainability.

4 – The future of CSP and PV

All the considerations and developments of the past decade have shaped the idea and potential use of CSP, introducing CSP+ plants as the new frontier for solar energy. Luque and Martmh. [62] proved in 1999 that hybrid systems integrating PV and CSP can be significantly more efficient – ideal efficiency values of 61.7% for hybrid systems, 40.0% for CSP, and 40.7% for photovoltaic – showcasing the potential of this hybridization. However, the focus on this

idea emerged again in the recent years, with Vossier, Verne [63] proposing two models for the hybrid system as shown in Figure 12. The first design is the “PV Mirror” (PVM) and integrates PV cells on the surface of CSP mirrors, allowing operations with the PV technology for photons in the optimal bandwidth and redirecting photons outside of this range to the CSP concentrator. The second design is called “PV topping” (PVT), where PV cells are integrated into the CSP’s thermal receiver and behave both as the thermal receiver and PV converter.

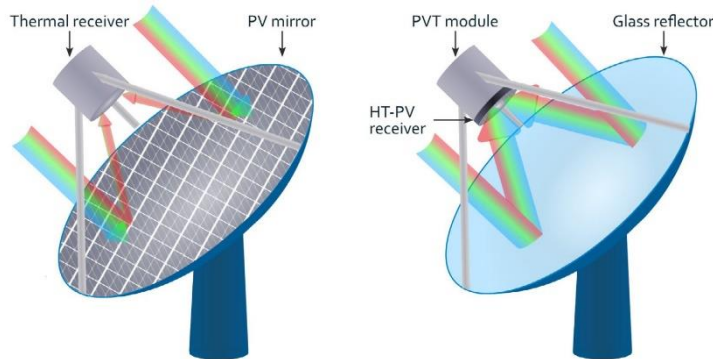


Figure 12: Diagram of the two configuration of hybrid systems (PVM and PVT) applied to a Parabolic Dish CSP module [63].

The study shows that the PVM approach integrated into the PTC technology has the highest power improvement relative to its CSP “equivalent” (without the PV cells). However, the improvement is strictly linked to the photovoltaic cell performance and thus with cell quality. The power balance is easily achieved using the PVM system, while in PVT the range of operating temperatures across which this balance exists is very narrow. Overall, the economic benefit outlined by Vossier, Verne [63] strongly varies based on the PV technology used and on the optimisation strategy followed, almost always approaching the CSP equivalent plant’s economic outline. Sumayli, El-Leathy [64] study the effect of hybrid systems in two cities in Saudi Arabia: Riyadh and Tabuk. The targeted capacity factor for the configurations modelled is 79%, and three main hybrid systems are assessed and compared to an equivalent CSP PT standalone: horizontal PV trackers combined with CSP (with CSP/PV capacity ratio of 1:1), a winter-optimized PV tilt (increasing the winter performance), and an oversized PV which redirects excess energy to the TES. The models used are the NREL’s System Advisor Model (SAM) for the simulation of CSP and PV systems and an economic analysis for the LCOE values. The costs and data for the economic assessment were derived from the latest dataset available from Saudi Arabia and the NREL, and by “optimization” is meant the process of achieving high capacity factors at a low LCOE. The study is very recent, and the results are meaningful: the solar multiples calculated are 6 and 3.5 for the standalone CSP system in Riyadh and Tabuk respectively, and they drop to 2.9-1.95 and 1.85-1.6 in the hybrid systems for the two cities. An interesting result is also the LCOE variation: in Riyadh, it reduced from \$0.177/kWh to \$0.140/kWh after hybridization, while in Tabuk it’s only minorly

affected and goes from \$0.137/kWh to \$0.134/kWh in the best case. Adding PV, in fact, reduces the solar field size required for CSP, decreasing costs. This shows how combining CSP's high thermal storage capacity with PV's low-cost electricity generation can significantly reduce LCOE and how hybrid systems make CSP technology viable for locations with varying solar resource profiles, effectively enlarging the suitable sites for solar energy generation. From the analysis, it also emerges that hybridization makes particular sense in locations with a higher fraction of diffuse sunlight – Riyadh – while in areas with a high DNI – Tabuk – standalone CSP is similar in performance and costs to hybrid systems. The location still plays a crucial role in defining the best strategy for hybrid systems: for example, excess power conversion to TS is more effective in Riyadh (around 20% reduction of LCOE) than in Tabuk (around 3% reduction in LCOE). Finally, the results also show how hybrid systems met the 45 MW baseload requirement year-round. The main findings of the paper then are that hybrid CSP-PV systems can significantly improve the cost-effectiveness of solar baseload generation, location-specific strategies are critical, oversizing PV for thermal storage integration is advantageous in regions with higher diffuse sunlight, and that hybrid configurations can support the transition to a fully renewable energy mix while maintaining high capacity factors. Costa, De Regel [65] dig deeper into the environmental and economic sides of CSP+ (hybrid) plants, with examples applied to plants in Belgium and Spain. The design of such plants consists of CSP PT technology combined with bifacial PV cells. The solar spectrum is split by a dichroic mirror, directing light rays to PV or CSP based on the wavelength. The study is carried out in three main scenarios: Scenario I (SCI) involves electricity-only production from the system, Scenario II (SCII) combines electricity and heat production, and Scenario III (SCIII) is a conventional PV and Concentrated Solar Thermal systems. To evaluate the environmental impacts, Costa, De Regel [65] used Life Cycle Assessment based on ISO 14040/44 standards. While for economic costs the metrics were LCOE and LCOH. It emerges that the main environmental benefits are the significant reduction of greenhouse gas emissions of the CSP+ plant compared to standalone CSP or PV and the PV cells' reduction of impacts to 51% of the total (down from 87% of standalone PV). The general reduction in climate change impacts is 83% for electricity and up to 97% for heat production (SCII), with values of 17 kg CO₂eq/ MWh in Ostend and 6 kg CO₂eq/MWh in Seville for SCII. On the economic side, SCII achieved the lowest LCOE (15% to 70% lower than benchmarks) and LCOH (up to 41% less compared to alternatives in Spain) values. Even this study confirms that combining CSP and PV enhances efficiency by optimally allocating the solar spectrum between technologies and that the use of bifacial PV cells improved energy

yield from diffuse sunlight. Moreover, the flexibility and modularity of hybrid systems allow for adaptability to various climatic conditions.

5 – Use of Generative AI

I utilized Generative AI tools, including ChatGPT and Adobe Acrobat AI Assistant, to support me in the writing of this literature research. Each tool has been used differently and therefore their contributions varied in effectiveness.

ChatGPT:

ChatGPT (4o-mini) was useful during the initial phase of structuring the research. After providing the research prompt and key background information, I tasked it to generate a report structure aligned with the rubric requirements. The output was valuable in generating a logical starting structure. However, two main issues were encountered: key points were sometimes omitted when adjusting the structure, and the outputs often lacked mutual exclusivity and clear organization, leading to repetitions.

Adobe Acrobat AI Assistant:

Adobe Acrobat AI Assistant excelled in summarizing, clarifying, and extracting information from academic papers. It delivered clear, concise responses to targeted prompts, particularly when analyzing specific chapters. This made it highly effective for processing extensive documents where deep analysis wasn't required.

Reflections:

The solar energy topic has been “trending” in recent years, making the results more unreliable due to the lack of traceability of GenAI sources, limiting the tools' usefulness for creating up-to-standard content from scratch. Additionally, an important note is that I have used ChatGPT since its release in November 2022 which has led it to adapt to my question style, introducing bias into its responses to try and give me “satisfaction”. In summary, GenAI is most effective for rewriting, summarizing, or extracting information from inputs provided by the user but is less reliable for original content creation due to personalization biases, unverified sources, and limited contextual understanding. Overall, generative AI served as an assistant but could not replace critical analysis and in-depth understanding of the topic.

Conclusions

This literature review has investigated the causes and circumstances that potentially influenced the recent development of Concentrating Solar Power and its impact on the renewable energy market, with a specific focus on the comparison with Photovoltaics. By

integrating and analysing the literature, it is possible to conclude that CSP has a very low impact on the current renewable energy market and has been significantly outpaced by PV in the solar industry. The main causes identified are costs, technical differences, and socio-political attitudes.

CSP plants have seen a decline in costs in the last 10 years, but the LCOE has plateaued at around 0.12 USD/kWh from the COVID-19 period onwards. In comparison, LCOE for PV has significantly declined, mainly driven by modules' cost reductions, and stands now at 0.044 USD/kWh. The LCOE is the main driver of the discrepancy in the market impact between PV and CSP, with the latter being significantly penalised by this difference.

Other important factors are the land suitability and the scalability of the technologies. CSP has generally stricter requirements, particularly for DNI and terrain slope values. These constraints make several "transition" areas – such as the lands bordering deserts or the south of Europe (excluding Spain) – unsuitable for CSP but suitable for PV, where this technology gains an incredible advantage. CSP is also hardly scalable and, regardless of the plant type, the land requirements are important, while PV benefits greatly from its modularity and can be deployed in several circumstances. In addition to higher land requirements, CSP plants require also a greater amount of water, mainly for cooling. While this field is being improved by trying to switch to air-cooling, it remains a strong discriminant for arid areas with few and far water reservoirs.

The policies and investments worldwide have picked PV as the betting technology but more and more countries like Morocco and China are starting to include CSP projects in their Renewable energy transition plans. Up until now, though, PV has been extremely popular and known, leading to a general preference and acceptance feeling compared to CSP which was seen as a more niche technology.

Finally, the strength of CSP technologies compared to PV is the energy storage system. While PV is usually coupled with batteries, CSP relies on Thermal Energy Storage which is generally cheaper, lasts longer, and minimally impacts the cost. However, TES has been implemented in CSP for some years and it doesn't seem sufficient to close the market gap with PV. Thus, new options are arising for CSP: hybrid PV and CSP plants can enhance performance, reduce operating costs, and tackle the pressing issue of dispatchability with integrated TES.

Therefore, CSP's market impact is now minimal for economic, technical, and social reasons but could gain momentum thanks to the development of hybrid solar plants.

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